

BREAK INTO AEROSPACE

Space Telecommunications Engineering

Subdomain Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Space telecommunications is undergoing the most consequential structural transformation in its history. The geostationary arc that defined the industry for fifty years now coexists with LEO broadband megaconstellations operating thousands of satellites simultaneously, and the defence sector is reshaping its satellite communications architecture around proliferated LEO for resilience. The technical demands span RF link engineering, antenna design, propagation modelling, payload design, orbit mechanics, spectrum management and interference coordination, ground segment architecture, and software-defined radio and network virtualisation.

Engineers in demand across a varied employer landscape: legacy GEO operators and satellite manufacturers (SES, Intelsat, Eutelsat, Telesat, Viasat, Hughes, Airbus DS, Thales Alenia Space, Boeing Space, Maxar Technologies), LEO megaconstellation operators (SpaceX Starlink, Amazon Kuiper, OneWeb/Eutelsat, Telesat Lightspeed), government and defence SATCOM (US Space Force MILSATCOM, ESA ARTES, UKSA, Inmarsat Government, ViaSat Government Systems, Kratos, General Dynamics Mission Systems), specialist subsystem suppliers (Teledyne Paradise Datacom, Comtech EF Data, iDirect, Newtec/ST Engineering), and new-space direct-to-device players (AST SpaceMobile, Lynk, SpaceMobile).

PART I · WHAT THE FIELD ACTUALLY LOOKS FOR

By Level

- **Intern / Graduate** — At entry level, space telecom employers are not primarily screening for satellite-specific knowledge — they are screening for RF fundamentals, signal processing intuition, and the ability to reason about power, noise, and bandwidth as a coupled system of constraints. The question that most reliably distinguishes strong from weak graduate candidates is not “what is a transponder?” but “if I increase my transmit power by 3 dB on a satellite link, what happens to the available margin, and under what conditions does doubling the transmit power not improve the link at all?” — a question requiring understanding of noise-limited versus interference-limited operation that separates students who have internalised the physics from those who have memorised the equations.

What distinguishes graduate candidates is evidence of hands-on RF engagement beyond coursework: a thesis involving RF link analysis, participation in a student satellite programme (CubeSat radio design, ground station operation), amateur radio licensing, or a software-defined radio project involving real signal reception, demodulation, or analysis. The field is one where practical RF experience is genuinely rare among graduates and is immediately recognisable to practising engineers.

- **Mid-Level (Engineer, Senior Engineer, ~3–8 years)** — At mid-level, space telecom interviews become system-specific and trade-off oriented. Questions shift from “explain the link budget” to “we have a GEO HTS satellite with 100 spot beams, Ka-band downlink, and a forward link that is interference-limited in certain beam clusters — walk me through how you would identify which beams are interference-limited and what payload or network-level changes would recover margin” or “our LEO constellation uses optical inter-satellite links — what are the pointing acquisition and tracking requirements that determine the ISL link margin budget, and what drives the closed-loop tracking bandwidth?”

The specific context driving mid-level hiring in 2025–2026: Starlink is continuously expanding its Gen2 constellation and integrating direct-to-cell capability; Amazon Kuiper is entering commercial service; OneWeb is integrating into Eutelsat’s combined LEO/GEO network; SDA’s Transport Layer is being contracted to Northrop Grumman, Lockheed Martin, and Rivada Space Networks; AST SpaceMobile’s commercial BlueBird constellation is entering operational service with mobile network operator partners; and ESA’s ARTES Advance programmes continue to fund next-generation HTS payload and optical ISL development.

- **Senior (Principal Engineer, Technical Lead, Fellow)** — At senior level, space telecom employers look for engineers who can define the technical architecture of a complete communication system — from satellite payload through ground segment through network management — and who can make defensible trade-offs when the requirements are in tension and the answer requires integrating RF physics, orbital dynamics, spectrum regulation, and network engineering simultaneously. The senior interview is a peer discussion with technical leadership, typically structured around a technical presentation of past work followed by a deep-dive discussion on a current challenge the organisation faces.

Senior roles at satellite manufacturers, operators, and government programme offices require engineers who have taken a complete communication system from requirements through launch and into operational use — and who have experienced the difference between what the pre-launch RF model predicted and what the on-orbit system delivers.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

Domain	Why It Signals Depth
Link Budget Analysis	Engineers who can work through a complete link budget — in both forward and return directions, including uplink and downlink contributions for a bent-pipe transponder — without reference to a template are demonstrating the baseline quantitative competence every space telecom role requires. The failure mode to avoid: treating the link budget as a spreadsheet to fill in rather than a physical argument to construct.
Antenna Fundamentals — Phased Arrays	The relationship between aperture area and gain ($G = 4\pi A_e/\lambda^2$), gain and beamwidth, and the consequence of pointing error on link margin are quantitative relationships tested directly. Phased array fundamentals — element spacing, grating lobe conditions, electronic beam steering, and scan loss — are increasingly tested as LEO systems rely heavily on phased array terminals and payloads.
Satellite Payload Architecture	Legacy bent-pipe transponders are transparent to the signal; digital transparent processors (DTPs) and on-board processors (OBPs) introduce demodulation and routing capability enabling flexible resource allocation. Engineers who understand what a DTP enables that a bent-pipe cannot — and what an OBP adds beyond a DTP — are demonstrating payload engineering depth that is specifically valued.
Spectrum Management and ITU Coordination	Engineers who understand what the ITU coordination procedure requires, what the epfd (equivalent power flux density) limits for LEO systems mean in practice, and how interference is characterised and mitigated in shared spectrum are valuable across operators, regulators, and spectrum management consultancies.
Modulation, Coding, and ACM	DVB-S2X defines the state-of-the-art MODCOD set for broadcast and interactive broadband. Understanding what E_b/N_0 threshold each MODCOD requires, how ACM exploits channel variability to maximise spectral efficiency, and what the capacity gain of ACM over fixed coding is in a rain-impaired Ka-band link is the technical vocabulary of commercial HTS system design.
Rain Fade and Propagation (Ka/V-band)	The ITU-R P.618 rain attenuation prediction model, the concept of availability as a link design parameter, and the mitigation strategies — power control, ACM, site diversity using geographically separated gateways — are baseline technical knowledge for any Ka-band system engineer.
Optical Inter-Satellite Links (OISLs)	Starlink's Gen2 constellation uses OISLs; Amazon Kuiper's architecture includes them; ESA's HyDRON programme and SDA's Transport Layer both incorporate optical links. The pointing, acquisition, and tracking (PAT) challenge at orbital velocities with sub-microradian accuracy requirements is specifically tested as OISLs enter operational deployment.
Software-Defined Radio and Signal Processing	Engineers who can implement a complete receiver chain — matched filter, timing recovery, carrier recovery, symbol detection, FEC decoding — in Python or C++ using real-sampled IF data are demonstrating the engineering depth that separates system architects from signal processing engineers.

How Space Telecom Interviews Differ from Generic RF or Aerospace Roles

- **The link budget is mandatory and quantitative.** Every technical interview in space telecom — at every level, at every employer type — will include a link budget discussion. Candidates who cannot engage with link budget mathematics in the interview — not just describe the concept but work through the arithmetic — will not progress at the technical stage of any serious space telecom process.
- **Spectrum is a constrained resource and regulatory literacy matters.** Space telecom design decisions are constrained by frequency assignments, orbital slots, coordination agreements, and ITU Radio Regulation limits that are not negotiable. An HTS operator cannot simply choose to transmit at higher power to improve a link — they are constrained by the EIRP and epfd limits in their coordination agreements and ITU filings. Engineers who treat spectrum as a physical given rather than a regulated allocation will make design errors that are not recoverable after the frequency filing is submitted.
- **The modality distinction between GEO and LEO is now deep.** An engineer who has spent a career in GEO SATCOM and is interviewing for a LEO megaconstellation role faces a genuinely different set of technical problems: inter-satellite link design, Doppler management, frequent handover, constellation maintenance, and a manufacturing and operations scale that is orders of magnitude larger than GEO satellite programmes. Interviewers will probe whether a candidate from one architecture genuinely understands the challenges of the other.
- **New-space hiring is faster and more data-oriented than legacy programmes.** Amazon Kuiper, SpaceX Starlink, AST SpaceMobile, and OneWeb hire engineers on timescales of weeks, not months, and use data-oriented technical screens — link budget problems, RF analysis exercises, sometimes coding challenges — rather than the multi-stage structured interview panels typical of Airbus, Thales Alenia, or Boeing.
- **Security clearance gates a significant fraction of the market.** US Military SATCOM requires minimum TS clearance; much of NRO and NSA-adjacent SATCOM work requires TS/SCI. UK SKYNET, UK military SATCOM architecture work, and NATO SATCOM work require DV or equivalent. The commercial SATCOM market is largely uncleared, though Viasat Government Systems, SES Government Solutions, and Intelsat General each maintain cleared divisions.
- **On-orbit operations experience is specifically sought.** Engineers who have operated a live satellite system — who have managed a link performing below pre-launch predictions and diagnosed the cause from telemetry alone — have an operational perspective that no design role can substitute for. This experience is specifically probed at senior level.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern / Graduate

1. Build and Work Through a Complete End-to-End Satellite Link Budget — Forward and Return, Including All Contributors 6–8 h

Skill: The foundational quantitative calculation of the entire field

Objective: The link budget is the single most important engineering calculation in space telecommunications. Every design decision — transmit power, antenna aperture, frequency band, modulation and coding scheme, margin allocation — is evaluated in terms of its effect on the link budget. A candidate who has never built one from scratch — not populated a template but derived each term from physical principles — has a gap that is immediately detectable in technical screening. The specific terms that reveal genuine understanding: why EIRP is in dBW and G/T is in dB/K and these add to give C/N_0 in dBHz; what the noise bandwidth of a receiver has to do with the bit rate; why

increasing the carrier frequency improves antenna gain for a given aperture size but also increases free-space path loss and rain attenuation. 6–8 h

Done Signal: You have built a complete link budget spreadsheet from scratch — not populated a template — for a Ka-band GEO forward link, including: transmit power and antenna gain to EIRP, free-space path loss at the link frequency and range, atmospheric and rain attenuation at the design availability, receive antenna gain and system noise temperature to G/T, and the resulting C/N₀, Eb/N₀, and link margin against the threshold Eb/N₀ for a specific MODCOD. You can explain every term in physical units, identify the two or three terms that most strongly drive the margin, and describe what changes if you increase frequency from Ku to Ka.

2. Understand the Frequency Bands Used in Satellite Communications — Why Each Is Used and What Constrains the Choice 3–4 h

Skill: Spectral band literacy as the context for every satellite system design decision

Objective: Satellite communications uses L-band (1–2 GHz), S-band (2–4 GHz), C-band (4–8 GHz), Ku-band (12–18 GHz), Ka-band (26.5–40 GHz), V-band (40–75 GHz), Q-band (33–50 GHz), and optical (1550 nm) — each with different propagation characteristics, rain fade susceptibility, antenna aperture requirements, regulatory status, and application suitability. L-band is used for mobile SATCOM because its large beamwidth is tolerant of non-stabilised terminals. C-band is used for broadcast because it is virtually unaffected by rain. Ka-band is used for HTS broadband because its high frequency allows small spot beams and high spectral reuse, at the cost of significant rain attenuation. Candidates who cannot explain why a specific band is chosen for a specific application are demonstrating spectral ignorance that is immediately limiting.

Done Signal: For each of L, S, C, Ku, Ka, and V bands: you can state the approximate frequency range, the primary satellite application, the dominant propagation impairment, and one specific satellite system or constellation that uses this band. You can explain why the transition from Ku-band to Ka-band for broadband HTS systems was driven by beam-size considerations, not just bandwidth availability. You can explain what optical inter-satellite links offer that RF ISLs cannot.

3. Understand Phased Array Antennas — Element Spacing, Beam Steering, Grating Lobes, and Scan Loss 3–4 h

Skill: The antenna technology defining the next generation of satellite ground terminals and payloads

Objective: Flat-panel phased array antennas are the dominant ground terminal technology for LEO broadband — Starlink's user terminal, Amazon Kuiper's terminal, and OneWeb's terminal are all electronically steered phased arrays. They are also increasingly used as satellite payloads for flexible multi-beam coverage. An engineer who does not understand the relationship between element spacing and grating lobe formation (element spacing greater than $\lambda/2$ produces grating lobes that alias energy into unwanted directions), why scan gain decreases approximately as $\cos(\theta)$ with off-broadside scan angle, and how a phased array achieves electronic beam steering without mechanical movement is missing the antenna fundamentals that underpin the most commercially active area of satellite terminal design.

Done Signal: You can explain why $\lambda/2$ element spacing is the standard for phased arrays and what happens at the first grating lobe if spacing exceeds λ . You can describe how electronic beam steering is achieved by applying a progressive phase shift across elements and compute the phase shift required to steer the beam to a given angle. You can explain what scan loss means in terms of effective aperture, and state approximately how many dB of gain a 60° off-broadside scan costs. You can describe one advantage and one limitation of a phased array compared to a mechanically steered parabolic dish for a LEO terminal application.

4. Prepare Behavioural Stories Connecting Technical Work to System Performance or a Delivered Outcome 2–3 h

Skill: STAR-format stories with the quantitative outcome emphasis space telecom employers probe

Objective: Space telecom interviewers specifically probe problem-solving under constraint — situations where a link margin was insufficient, an interference source was unexpected, a schedule pressure forced a design trade, or a system performed below prediction and required on-orbit diagnosis. The stories that are weakest in this field end with “the system met spec.” The stories that are strongest end with “the system was 1.5 dB short of the required margin on the eastern beam cluster, we traced it to an underestimate of rain attenuation at the gateway latitude, and we recovered margin through a combination of adaptive coding and gateway site diversity that we had not originally planned for.”

Done Signal: You have four STAR stories covering: a link or system performance problem you diagnosed using data, a cross-disciplinary collaboration (RF engineering and software, payload and ground segment, or systems and operations), a design trade-off you made under a specific constraint, and a failure or underperformance you identified and corrected. At least two stories include specific numerical performance metrics.

5. Build Working Knowledge of Modulation and Coding for Satellite Links — DVB-S2/S2X, MODCOD Thresholds, and ACM 3–4 h

Skill: The spectral efficiency layer that determines satellite link capacity

Objective: DVB-S2X defines the modulation and coding scheme options for modern satellite broadband forward links — 116 MODCODs spanning from QPSK 2/9 (for deep fades) to 32APSK 9/10 (for excellent channels), each with a specific E_b/N_0 threshold. Understanding what ACM does — dynamically selecting the highest-order MODCOD the current channel condition supports, maximising throughput during clear sky and gracefully degrading during rain fade — and why this is a qualitative improvement over fixed coding is the operational vocabulary of HTS system design. Candidates who cannot connect a C/N_0 value to a specific MODCOD, explain what the quasi-error-free threshold means, or describe the capacity gain that ACM provides are missing the layer of the link budget that determines what the system actually delivers to users.

Done Signal: You can explain what ACM does and why it increases average throughput relative to fixed coding. You can state approximately what E_b/N_0 threshold a QPSK 3/4 or 8PSK 2/3 MODCOD requires. You can describe the relationship between spectral efficiency (bits per symbol), required E_b/N_0 , and link margin — and explain why higher-order modulation is not always preferred even when the link margin would theoretically support it. You know what DVB-RCS2 specifies and how it differs from DVB-S2X in terms of channel direction and access method.

6. Get Hands-On with a Software-Defined Radio — Receive, Demodulate, or Analyse a Real Satellite Signal 4–6 h

Skill: Practical RF experience as the field’s most differentiating portfolio signal for graduates

Objective: Practising RF engineers identify hands-on signal experience immediately. An RTL-SDR dongle costs approximately £20 and a suitable antenna can be built for less. With it, a candidate can receive NOAA weather satellite APT signals (FM-modulated at 137 MHz), Inmarsat STD-C safety messages (BPSK at 1.5 GHz with a patch antenna), ADS-B aircraft transponders (1090 MHz), or AIS vessel transponder signals (161–162 MHz). Processing any of these — receiving the signal, performing FFT analysis to identify the carrier, applying the appropriate demodulation, and decoding the payload content — demonstrates practical RF competence that no coursework can substitute for.

Done Signal: You have received at least one real satellite or aircraft radio signal using an RTL-SDR or equivalent, performed spectral analysis of the received signal, applied appropriate demodulation

4–6 h

(FM for APT, BPSK for Inmarsat STD-C, or equivalent), and successfully decoded content from the received signal. You can describe the signal structure, the reception challenges you encountered (Doppler on LEO satellites, polarisation mismatch, interference sources), and what you would do differently to improve receiver performance.

Mid-Level (~3–8 years)

All graduate tasks still apply. Behavioural stories must come from professional contexts involving real satellite systems, real spectrum environments, or real operational scenarios.

4–6 h

7. Master Rain Fade Analysis and Gateway Site Diversity Design for Ka/V-Band HTS Systems

Skill: The dominant impairment in modern HTS systems and its primary mitigation strategy

Objective: Rain attenuation at Ka-band and V-band is the primary link availability challenge for modern HTS gateways. Gateway links must achieve very high availability — typically 99.9% or better — because a gateway outage affects all users served by that gateway's capacity. Achieving 99.9% availability at Ka-band in a high-rain-rate region may require margins exceeding 15 dB — prohibitively expensive if achieved by link margin alone. Gateway site diversity — placing two geographically separated gateways in the same beam — recovers most of this margin at the cost of a second gateway site and the switching architecture. Engineers who can compute a site diversity gain using ITU-R P.618, select appropriate separation distances for a target joint probability of fade, and describe the switching architecture needed to exploit it are demonstrating the applied propagation engineering competence that Ka-band system design requires.

Done Signal: You can work through a Ka-band gateway site diversity calculation: using ITU-R P.618 to estimate single-site rain attenuation exceeded for a given percentage of time, applying a site diversity gain model to estimate the joint attenuation at a specified separation distance, and deriving the effective link availability improvement. You can describe the gateway switching architecture — automatic switchover, traffic re-routing, and beam illumination management — that operationalises the diversity gain. You can state what determines the optimum gateway separation distance and why co-located gateways provide no diversity gain.

4–5 h

8. Understand Satellite Payload Architecture in Depth — Bent-Pipe, Digital Transparent Processor, and On-Board Processor

Skill: The technology transition reshaping GEO HTS payload design

Objective: Satellite payload flexibility is the central commercial challenge for GEO HTS operators, whose fixed beam coverage and power allocation cannot adapt to demand patterns that shift across geography and time. The digital transparent processor digitises the received spectrum, routes frequency channels between beams digitally, and re-transmits without full demodulation — enabling dynamic power and bandwidth allocation across beams. On-board processors go further, demodulating, decoding, switching, re-encoding, and re-modulating traffic, enabling full store-and-forward operation and interference cancellation. The trade between these architectures — in terms of power consumption, mass, flexibility, complexity, and failure mode — is a live design discussion in every GEO satellite development programme.

Done Signal: You can explain what a digital transparent processor does physically — where the ADC and DAC interfaces are, what digital channelisation does, and how beam-hopping is implemented in a DTP payload. You can describe what an on-board processor adds — the demodulation, decoding, and re-encoding stages — and explain what interference cancellation capability an OBP enables that a DTP does not. You can name one operational GEO satellite with a DTP or OBP pay-

load and describe its published capacity and flexibility characteristics.

4–5 h

9. Prepare a Technically Specific Story About Diagnosing and Resolving a Satellite Link or Interference Problem — Using Real Data

2 h

Skill: Operational engineering judgement under imperfect information

Objective: Satellite communications systems encounter interference, anomalous link performance, and unexpected propagation effects that are not predicted in the pre-launch model. The ability to diagnose these from telemetry, spectrum analysis, and system performance data — without access to the satellite or the ability to stop the traffic — is the operational engineering skill most valued by satellite operators and network operations centres. Engineers who have diagnosed a real interference event, a link underperformance relative to prediction, or a payload anomaly have an operational depth that no pre-launch design role can provide.

Done Signal: You have a story about a satellite link or interference problem you diagnosed and resolved using operational data. The story names the symptom (degraded C/N_0 , increased BER, intermittent outage, interference-induced capacity reduction), the data you examined (spectrum plots, telemetry trends, performance monitor logs), the hypothesis you formed and tested, the root cause you identified, and the operational or engineering response you implemented. The story comes from professional work and ends with a specific outcome.

10. Build Working Knowledge of LEO Megaconstellation Architecture — Walker Orbits, Inter-Satellite Links, Doppler Management, and Handover

4–5 h

Skill: The dominant commercial growth architecture in space telecommunications

Objective: LEO broadband megaconstellations — Starlink at approximately 6,000 operational satellites, Amazon Kuiper at 3,236 satellites, OneWeb at 648 satellites, and others in development — represent the dominant commercial growth trajectory in space telecommunications. Their architecture differs fundamentally from GEO: each satellite has a limited field of view requiring frequent handover of user terminals, Doppler shifts of tens of kilohertz require active compensation in the modem, optical inter-satellite links enable ground-station-independent routing, and the constellation design problem determines coverage, capacity, and launch cost simultaneously. Engineers who understand Walker delta constellation geometry, can compute the number of satellites required for continuous coverage at a given minimum elevation angle, and can describe the Doppler compensation approach for a Ka-band LEO link at 550 km altitude are demonstrating the systems engineering depth that LEO constellation operators require.

Done Signal: You can compute the approximate number of satellites required for continuous single-coverage of the Earth at 550 km altitude and 53° inclination for a 25° minimum elevation angle constraint. You can explain what a Walker delta constellation specifies (T/P/F notation), why inclination determines the latitude of coverage gaps, and why Starlink's shell at 53° inclination provides good coverage of mid-latitudes but incomplete coverage of the poles. You can describe the Doppler shift magnitude at 550 km altitude for a Ka-band downlink at a terminal at 45° elevation, and explain how a terminal's tracking loop must accommodate it.

11. Understand ITU Radio Regulations and Spectrum Coordination — Orbital Filing, efd Limits, and the GEO/non-GEO Interference Regime

4–5 h

Skill: The regulatory framework that constrains every satellite system design and operation

Objective: Satellite frequency assignments are not engineering choices — they are administered rights derived from ITU coordination procedures, filed years before launch, and constrained by the

4–5 h

regulatory limits applicable to the frequency band and orbital regime. The ITU's epfd (equivalent power flux density) limits for non-GSO systems — defined in Article 22 of the Radio Regulations — are the specific regulatory constraint that Starlink, Kuiper, OneWeb, and other LEO broadband systems must satisfy when operating in frequency bands shared with GEO services. Understanding what epfd measures (the aggregate interference power spectral density that a non-GSO system produces into a GEO earth station), why the limits are defined as a statistical threshold, and how a LEO operator manages its constellation to comply — through satellite power reduction in specific orbital geometries — is the regulatory literacy that satellite operators and spectrum management engineers require.

Done Signal: You can explain what an ITU advance publication, coordination request, and notification represent in the satellite filing procedure, and approximately how many years before launch each must be submitted. You can describe what an epfd limit means physically — the aggregate interference power flux density from all visible non-GSO satellites into a hypothetical GEO earth station's reference antenna — and why it is defined as a percentage-of-time threshold rather than an absolute limit. You can describe at least one operational technique that a LEO constellation operator uses to manage epfd compliance as satellites pass near the GEO arc.

4–5 h

12. Understand HTS Ground Segment Architecture — Gateway Design, Feeder Link, and Network Resource Management

Skill: The ground segment as the operational control layer of satellite capacity

Objective: An HTS satellite delivers capacity to a service area only if the ground segment can absorb and route that capacity efficiently. Gateway design — the number of gateways, their geographic placement, the feeder link EIRP and G/T, the hub architecture, and the baseband processing capacity — determines how much of the satellite's payload capacity reaches end users. Network resource management systems allocate bandwidth, power, and frequency resources dynamically across potentially millions of user terminals, implementing QoS policies, fairness rules, and priority handling. Engineers who understand how a gateway's feeder link interfaces to the satellite payload, how capacity is allocated dynamically to user beams, and what happens during a gateway site diversity switchover are demonstrating the ground segment depth that HTS operators and ground segment integrators require.

Done Signal: You can describe the functional architecture of an HTS gateway: the antenna and modem chain for the feeder uplink, the IF distribution and baseband processing for user traffic, and the interface to the network management system. You can explain what a frequency plan for a multi-beam HTS satellite looks like — how uplink and downlink frequencies are assigned to beams to maximise frequency reuse while managing co-channel interference. You can describe how a network management system implements dynamic bandwidth allocation in a TDMA return link, and what determines the maximum number of terminals a single gateway can support.

4–6 h

13. Prepare for a Live or Take-Home RF Engineering Problem — Link Budget, Interference Calculation, or Coverage Analysis

Skill: Producing correct quantitative RF analysis under interview conditions

Objective: Space telecom employers at new-space companies and specialist RF houses use take-home or live technical problems involving RF calculations. A typical problem might provide a satellite EIRP, orbit altitude, frequency, receive antenna parameters, and a noise temperature, and ask for the C/N_0 , the achievable bit rate at a specific MODCOD, and the effect on link margin of increasing the rain fade by 3 dB. The candidate who has built link budgets regularly completes this in 20 minutes; the candidate who has seen the concept but not worked the numbers in years will struggle with unit conversions, the decibel arithmetic, and the sign convention for attenuation terms.

4–6 h

Done Signal: You can work through a complete Ka-band forward link budget in under 20 minutes, given the transmit parameters and receive parameters, arriving at C/N_0 in dBHz and link margin in dB, with all unit conversions correct. You can compute the interference C/I contribution from a specified interferer EIRP at a given angular separation from the desired satellite, using the receive antenna sidelobe gain pattern. You know the approximation for free-space path loss ($FSPL \approx 92.4 + 20 \log(f_{\text{GHz}}) + 20 \log(d_{\text{km}})$ dB) and can apply it without looking it up.

Senior Level (Principal Engineer, Technical Lead, Fellow)

All mid-level tasks still apply. Link budget knowledge must extend to system-level margin allocation, on-orbit calibration, and the propagation of uncertainty through the budget. Payload knowledge must extend to implementation trade-offs, not just architectural description.

4–6 h

14. Prepare to Walk Through the Complete Development of a Satellite Communication System — from Requirements Through On-Orbit Commissioning

Skill: End-to-end system ownership at the depth a senior space telecom engineer must demonstrate

Objective: Senior space telecom engineers at satellite manufacturers, operators, and government programme offices are expected to own technical threads from system requirements through satellite design through ground segment specification through launch and early orbit operations (LEOP) through in-service performance validation. For an HTS satellite, this means being able to discuss the beam pattern trade study, the payload power budget, the link budget allocation at Beginning-of-Life and End-of-Life, the on-orbit commissioning approach (beam pattern verification, EIRP and G/T measurement, interference characterisation), and the performance monitoring approach for the mission lifetime. The on-orbit commissioning experience — specifically the process of discovering and diagnosing discrepancies between the pre-launch RF model and the first measured on-orbit performance — is the definitive seniority signal in this field.

Done Signal: For one satellite communication system or payload you personally owned the RF or systems engineering for — not contributed to, owned — you can discuss without notes: the key system requirement that drove the design, the main link budget trade-off made, the approach to on-orbit commissioning RF verification, the most significant discrepancy found between pre-launch prediction and on-orbit performance, and how it was diagnosed and resolved. You can quantify the discrepancy and the resolution in engineering units.

4–5 h

15. Prepare a Technically Grounded Position on One Active Frontier in Space Telecom — Direct-to-Device, Optical ISLs, or Software-Defined Satellite Payloads

Skill: Forward-looking technical awareness at senior engineering depth

Objective: Space telecommunications in 2025–2026 has several genuine technical frontiers. Direct-to-device SATCOM — connecting standard mobile handsets directly to LEO satellites without specialised terminals, as being pursued by AST SpaceMobile (BlueBird constellation), Starlink's direct-to-cell service, and others — requires overcoming a link budget challenge of approximately 30 dB relative to a cellular base station link, demanding very large satellite apertures, aggressive frequency reuse, and receiver sensitivity governed by 3GPP specifications the satellite operator cannot change. Optical ISLs face the pointing, acquisition, and tracking challenge at orbital velocities with sub-microradian accuracy requirements. Software-defined satellite payloads — fully reconfigurable in orbit via ground-uploaded software — face the question of what reconfigurability is commercially worth against its mass, power, and cost premium.

Done Signal: You have a prepared 3-minute technical discussion covering one specific engineering

4–5 h

challenge on one frontier — named specifically: the link budget challenge for direct-to-device at the 3GPP NR NTN power class limits; the PAT (pointing, acquisition, and tracking) bandwidth and control system design requirements for OISLs at 550 km orbit; or the interference coordination challenge of operating a software-defined multi-beam payload in dynamic frequency allocation mode within ITU filing constraints. Your position identifies a specific technical problem, describes the engineering approaches being taken, and states your assessment of what is and is not yet solved.

4–5 h

16. Prepare to Lead a Technical Discussion on a Satellite System Architecture Trade-Off for a Specific Application Scenario

Skill: Architectural decision-making at senior engineering speed

Objective: Senior space telecom engineers make architectural recommendations under competitive and schedule pressure, often with imperfect information. “A government customer requires assured satcom capability for a deployed force at northern European latitudes with 99.99% availability, resistance to jamming, and a 50 ms round-trip latency requirement — walk me through your architecture trade” requires reasoning through: GEO versus LEO (latency immediately eliminates pure GEO at 600 ms RTT), LEO military protected waveform availability, coverage at 60–70°N latitude (requiring high-inclination orbit or GEO supplemented by LEO), gateway diversity for availability, and the terminal class that can be fielded with military personnel. The senior interview tests whether you can structure this reasoning live, reach a specific recommendation, and defend it against challenge.

Done Signal: You have worked through at least three satellite communication architecture trade problems spanning at least two of: military assured SATCOM for a deployed ground force, broadband connectivity for a maritime fleet in the North Atlantic, direct-to-device emergency communications for disaster response in a region without cellular infrastructure, or high-throughput gateway connectivity for a remote oil and gas platform. For each, you state the requirements that most strongly constrain the architecture, the viable architectural options, your recommended architecture, and the key technical risk that would cause you to revisit it.

3–4 h

17. Understand the EHF and SHF Military SATCOM Architecture — AEHF, WGS, and the Protected Tactical Waveform

Skill: The technical architecture of the most survivable military SATCOM layer

Objective: Military satellite communications is architecturally distinct from commercial SATCOM in its anti-jam, anti-intercept, and nuclear-survivability requirements. The Advanced Extremely High Frequency (AEHF) system — three operational satellites at GEO, operating at EHF (44 GHz uplink, 20 GHz downlink) with crosslinks — provides protected strategic SATCOM with jam resistance derived from its small uplink beam and frequency hopping waveform. The Wideband Global SATCOM (WGS) system operates in Ka/X-band and provides wideband capacity without AEHF’s jam resistance. The Protected Tactical Waveform (PTW) is the US DoD’s standard for tactical SATCOM communications resistant to jamming and spoofing in contested environments. Senior engineers working in defence SATCOM are expected to understand the performance characteristics of these systems, their limitations, and the architectural direction of the SDA Transport Layer that supplements them.

Done Signal: You can describe, at an unclassified level, what distinguishes AEHF from WGS in terms of frequency, jam resistance, and target user community. You can explain what the Protected Tactical Waveform provides in terms of anti-jam margin and why the EHF uplink frequency contributes to it. You can describe what the SDA Transport Layer is intended to provide that AEHF and WGS do not — specifically in terms of proliferation for resilience, lower latency, and direct access for tactical users. You understand which aspects of your discussion would transition to classified detail in a cleared environment.

18. Build Quantitative Estimation Fluency for Space Telecom System Performance — Without Running a Simulation 4–5 h

Skill: Senior-speed system-level reasoning about link margin, capacity, coverage, and availability

Objective: Senior space telecom engineers are expected to sanity-check design proposals, estimate whether a proposed system can meet a requirement before committing to a design, and respond credibly to “does this close?” in programme reviews — in real time. The estimation problems are tractable from first principles: computing FSPL at a given frequency and range, estimating rain attenuation from ITU-R P.618 for a given elevation angle and target availability, estimating the capacity of a spot beam from its bandwidth and achievable MODCOD, estimating the number of gateways required for a given HTS throughput target, or estimating the Doppler offset at LEO for a given carrier frequency and satellite velocity. None of these requires a simulation — they require physical intuition and the ability to carry units through an estimate without a spreadsheet.

Done Signal: You have worked through five system performance estimation problems spanning at least two of: FSPL and link margin estimation at a given frequency and range, rain attenuation and availability estimation for a Ka-band gateway at a given latitude, spot beam capacity estimation from bandwidth and MODCOD, gateway count estimation for an HTS throughput target, Doppler offset estimation for a LEO Ka-band link. You state your assumptions before your numbers, identify the most load-bearing assumption, and arrive at an order-of-magnitude answer with stated uncertainty.

All Levels

19. Prepare a Specific Answer to “Why Space Telecom — and Why This Specific Segment or Application?” That Engages with the Field’s Current Moment 1–2 h

Skill: Technically specific motivation that distinguishes field-committed candidates

Objective: Space telecommunications is a field in genuine architectural transition. Candidates who describe their motivation in terms of “connecting the unconnected” or “advancing satellite technology” are giving answers that could apply to any space communications role in any decade. The expected answer in 2025–2026 engages with something specific: the direct-to-device challenge and what it requires technically; the SDA Transport Layer and what proliferated LEO means for military SATCOM resilience; the optical ISL deployment on Starlink and what it changes about routing and latency; the software-defined payload and what dynamic resource allocation enables for a GEO operator managing variable demand patterns; or the integration of non-terrestrial networks (NTN) into 3GPP 5G standards and what this changes about satellite system design.

Done Signal: You have a 90-second answer that names your target segment within space telecom (GEO HTS, LEO megaconstellation, military SATCOM, direct-to-device, or optical communications), identifies one specific active engineering challenge in it, explains what makes that challenge technically hard, and connects your engineering background to contributing to its solution. The answer cannot be given unchanged to a candidate for a generic satellite engineering or RF engineering role.

20. Map Your Technical Background to a Specific Employer Type, Segment, and Role — and Verify Security Clearance Eligibility 1–2 h

Skill: Precise self-positioning in an employer landscape spanning legacy GEO operators to new-space LEO startups to classified government programmes

Objective: The space telecom industry spans employers as different as a 50-person LEO direct-to-device startup moving at startup velocity and a 200-person MILSATCOM programme team at a cleared defence prime operating under export control and programme security requirements. An RF systems engineer at SpaceX Starlink is designing link budgets for a constellation that changes its architecture quarterly. An RF engineer at Thales Alenia Space is working within a structured GEO

satellite development programme that spans eight years from contract to launch. A spectrum engineer at SES navigates ITU coordination procedures on a five-year horizon. A signal processing engineer at L3Harris works on classified anti-jam waveforms for military terminals. These roles share a technical vocabulary but require different backgrounds, tolerate different communication styles, and offer very different career trajectories.

1–2 h

Done Signal: You can state in under 60 seconds: the space telecom segment you are targeting, the specific technical domain within that segment that is your primary strength (payload RF, link engineering, ground segment, network management, or spectrum), the programming and analysis tools you work in fluently, your security clearance status or eligibility, and two specific technical contributions your background makes to the target role. You have verified all of this against at least one specific vacancy.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	End-to-end link budget — forward and return, all contributors	All	6–8 h
2	Frequency bands — why each is used, propagation, application	All	3–4 h
3	Phased array antennas — element spacing, beam steering, grating lobes	All	3–4 h
4	STAR stories — link performance, interference, trade-off, failure	All	2–3 h
5	Modulation and coding — DVB-S2X, MODCOD thresholds, ACM	All	3–4 h
6	Hands-on SDR — receive, demodulate, or analyse a real satellite signal	All	4–6 h
7	Rain fade and gateway site diversity — Ka/V-band, ITU-R P.618	Mid+	4–6 h
8	Satellite payload architecture — bent-pipe, DTP, OBP, what each enables	Mid+	4–5 h
9	Link or interference diagnosis story — operational data, root cause	Mid+	2 h
10	LEO megaconstellation architecture — Walker, Doppler, handover, ISLs	Mid+	4–5 h
11	ITU Radio Regulations — filing, epfd limits, GEO/non-GEO regime	Mid+	4–5 h
12	HTS ground segment — gateways, feeder links, network resource management	Mid+	4–5 h
13	Live RF problem readiness — link budget and interference, under time pressure	Mid+	4–6 h
14	Complete satellite communication system — requirements through on-orbit	Senior	4–6 h
15	One active frontier — direct-to-device, optical ISLs, or SDR payloads	Senior	4–5 h
16	Satellite system architecture trade-off discussion — live, under challenge	Senior	4–5 h
17	Military SATCOM architecture — AEHF, WGS, PTW, SDA Transport Layer	Senior	3–4 h
18	System performance estimation — five problems, units carried	Senior	4–5 h
19	“Why space telecom — why this segment?” — technically specific	All	1–2 h
20	Segment / domain / clearance self-positioning	All	1–2 h

THE STRUCTURAL REALITY

Space telecommunications is the only aerospace subdomain where every design decision is simultaneously constrained by physics, by orbital mechanics, by international regulatory agreements, and by a commercial market that moves faster than the satellites it depends on. The engineer designing a Ka-band spot beam is not choosing a transmit power — they are choosing a position in a four-dimensional space defined by link margin physics, ITU coordination agreements, rain attenuation statistics, and the network architecture that must connect that beam to the terrestrial internet via a gateway that does not yet exist at the location the coverage model requires.

The field rewards engineers who can hold all of these constraints simultaneously without losing the physics. The RF engineer who does not understand the ITU filing consequences of their power level choice will create regulatory problems that cannot be resolved without fundamental redesign. The systems architect who does not understand the propagation statistics of the frequency band they have chosen will specify a gateway availability that the site diversity architecture cannot achieve.

This integration of RF physics, orbital mechanics, regulatory constraints, and network engineering is what distinguishes the strongest space telecom engineers from those who are strong in one or two of these dimensions and weak in the others. The satellites being designed and launched in 2025–2026 will still be operating in 2040. The frequency assignments being filed today will govern spectrum access for decades. Engineers who understand the full depth of what those choices require, and who can make them correctly under constraint, are among the most consequential practitioners in the space sector today.

BREAK INTO AEROSPACE
